

## Trøndelagkonkurransen

### Variety of Problems

1. The figure below shows two circumferences of centers  $O$  and  $O'$  tangent to each other.  $A$  in the larger circumference is collinear with  $O$  and  $O'$ . The chords  $AP$  and  $AQ$  are tangent to the inner circumference. If  $\angle QAP = 60^\circ$ , determine the ratio of the areas of the circles.

